

# LAURENT "ENZO" NTIBARIKURE

Engineering | Energy & Information Systems | Authoring

@ntilau@gmail.com

in ntilau

ntilau

Eindhoven, Netherlands

## SUMMARY

Experienced program management professional with a strong technical background in electronic and mechatronic systems development. Proven ability to lead cross-functional teams, manage complex multi-stakeholder initiatives, and drive educational and technology-focused programs from concept to implementation. Adept at translating technical requirements into actionable plans while maintaining strong organizational oversight in dynamic, international environments.

## EXPERIENCE

### Sr. Engineer - Design Enablement

**NXP Semiconductors**

03/2022 - Present

Eindhoven, Netherlands

- Managed cross-functional technical programs developing electromagnetic simulation flows for SoCs and antenna systems
- Coordinated with vendors and internal teams to enhance solutions for design and cloud platform integration requirements
- Led development of problem resolution automation scripts and utilized parametric reduced order models to improve program efficiency
- Served as technical liaison between engineering teams and stakeholders to ensure alignment on program objectives

### Technical Account Manager

**Ansys**

03/2020 - 03/2022

Utrecht, Netherlands

- Directed customer satisfaction programs through multi-disciplinary technical support and global presales activities
- Led cross-disciplinary teams to architect solution flows, focusing on requirements gathering and critical success factors
- Advised global sales and account engineers on technical initiatives and solution visions, managing stakeholder expectations
- Served as primary point of contact for complex technical programs spanning multiple product lines and industries

### Sr. Applications Engineer

**Ansys**

01/2019 - 03/2020

Utrecht, Netherlands

- Managed presales programs for electronic systems across industries, coordinating technical resources and customer engagements
- Led technical alignment programs with customers on automotive electrification and connectivity projects
- Organized and delivered consulting services and training programs to ensure successful technology adoption

### Lead EIC Engineer - Testing

**Baker Hughes**

07/2016 - 12/2018

Florence, Italy

- Directed the development of industrial, ATEX compliant, distributed data acquisition systems to streamline test execution from months to weeks
- Managed cross-functional teams to resolve 5+ electromagnetic interference issues in NPI validations and factory acceptance tests
- Led innovation programs resulting in patent disclosure for Structural Vibration Monitoring System and multiple customer impact awards

## EDUCATION

### Doctor of Philosophy - PhD, Electronics Engineering

**Università degli Studi di Firenze**

01/2011 - 02/2014 Florence, Italy

- Managed research programs in applied & computational electromagnetics, presenting 10+ papers
- Directed student thesis advisement programs and supported examination processes
- Developed high-frequency electromagnetics 3D finite element C++ code for domain decomposition and nonlinear formulations
- Architected and demonstrated methodology programs using Simulink, Comsol AC/DC and HFSS for electromagnetic compatibility in variable speed drive systems (GE/Nuovo Pignone collaboration)

### Master's Degree, Electronics Engineering

**Università degli Studi di Firenze**

10/2007 - 04/2010 Florence, Italy

- Managed Erasmus Placement/LLP framework thesis semester at Lehrstuhl für Theoretische Elektrotechnik, Universität des Saarlandes
- Developed and directed intrusive model order reduction programs in finite element analysis of antenna arrays
- Implemented Matlab code programs for 3000x matrix reduction and 300x speedup in antenna pattern computation

### Bachelor's degree, Electrical and Electronics Engineering

**Università degli Studi di Firenze**

01/2003 - 04/2008 Florence, Italy

- Directed thesis project at Laboratorio di Tecnologia per i Beni Culturali
- Developed and managed wireless connectivity programs for battery-powered avalanche rescue ground penetrating radar
- Executed design and assembly programs of ISM 868 MHz serial modem PCB prototype
- Achieved successful wireless system programs resulting in improved breath signal detection quality
- Served as Treasurer @ Student Union, managing financial and operational programs

### High School Diploma

**École Autonome Belge à Bujumbura**

09/1996 - 06/2002 Bujumbura, Burundi

- Coordinated activities across engineering, testing, and customer teams to ensure program deliverables met specifications
- Participated in extracurricular programs including Football, Basket, Volley

EIC Engineer - Testing

Baker Hughes

06/2013 - 06/2016 Florence, Italy

- Executed electrical, instrumentation and controls design activities for GE Oil & Gas NTI/NPI programs
- Contributed to rapid test execution programs through internal standardization and inventory optimization initiatives
- Collaborated with multidisciplinary teams to ensure uncertainty-assessed special instrumentation met performance validation requirements

Startup Co-Founder

Università degli Studi di Firenze

06/2012 - 06/2013 Florence, Italy

- Founded and managed ElectroSoft spin-off to provide efficient numerical codes for design problems
- Led team development and value proposition programs, pitching to business angels and potential investors
- Developed and managed business development programs targeting defense industry opportunities

TECHNICAL PROJECTS DEMONSTRATING PROGRAM MANAGEMENT SKILLS

LM9000: 65 MW aeroderivative gas turbine

Baker Hughes, a GE Company

06/2018 - 12/2018 Florence, Italy

- Contributed to conceptual instrumentation design programs and test rig erection initiatives
- Coordinated data acquisition architecture and control system design through technical and HAZOP review processes

Bearings loading test rig

Baker Hughes, a GE Company

05/2018 - 12/2018 Florence, Italy

- Directed conceptual design programs for novel bearings loading test rig aimed at validating bearings NTIs
- Managed investigation programs on EM coupling modeling methods between power systems and signaling cabling

Gas Turbine embedded SAW sensors

Baker Hughes, a GE Company

02/2018 - 12/2018 Florence, Italy

- Co-advised MEng thesis student programs with University of Florence on SAW sensors high frequency numerical modeling
- Managed accuracy control programs for temperature measurement initiatives

GE Aviation's Catalyst

Baker Hughes, a GE Company

06/2017 - 12/2018 Prague, Czech Republic

- Directed special instrumentation design programs and global technical reviews of FETT
- Coordinated with design and testing teams across Czech Rep., Poland, US, and Canada
- Managed >4 M\$ cross-P&L project programs and customer satisfaction initiatives

AWARDS

EMEA TAM of the Year | Ansys (01/2021)  
Award for customer and industry focus

CERTIFICATIONS

Professional Engineer,  
ID:ES2010277049000038  
Università degli Studi di Firenze

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## GE-6G PowerGen Module Structural Vibrations Monitoring System

[GE Oil & Gas](#)

📅 11/2015 - 03/2016

📍 Massa, Italy

- Directed development of ATEX-compliant distributed acquisition system for 150+ vibration measurement points
- Managed 40 kHz analog bandwidth, WiFi5 backhaul to storage, and real-time monitoring programs
- Designed and implemented 3D assemblies and BOM programs for timely delivery and cost optimization
- Introduced software-based velocity and displacement real-time computation programs from accelerometers
- Achieved direct \$150k savings through innovative lab processes and OPEX benefits

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## Finite Element Software (FES)

[Università degli Studi di Firenze](#)

📅 06/2011 - 02/2014

📍 Florence, Italy

- Managed open source framework development programs for "ad-hoc" simulation software
- Directed rapid implementation programs of novel numerical methods based on the Galerkin method
- Coordinated verification programs using meshes from commercial packages (Comsol, HFSS)

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## Phased array antennas model order reduction (MEng thesis)

[Universität des Saarlandes](#)

📅 10/2009 - 02/2010

📍 Saarbrücken, Germany

- Developed and managed intrusive model order reduction programs in finite element analysis
- Implemented Matlab code programs for 3000x matrix reduction and 300x speedup in antenna pattern computation

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## COURSES

### Command of the Message

[Ansys](#)

📅 06/2020

📍 Remote

### JAWS: Just Another Way of Selling

[Ansys](#)

📅 02/2020

📍 Paris, France

### HBM Dynamic Load Cells

[GE Oil & Gas](#)

📅 05/2016

📍 Bologna, Italy

### GE Crotonville: Delivering Customer Impact

[GE Oil & Gas](#)

📅 03/2016

📍 Munich, Germany

### GE Control Valves — Advanced

[GE Oil & Gas](#)

📅 09/2015

📍 Florence, Italy

### CE Marking Overview

[GE Oil & Gas](#)

📅 11/2014

📍 Florence, Italy

### Safety Integrity Levels

[GE Oil & Gas](#)

📅 10/2014

📍 Florence, Italy

### Electrification

[GE Oil & Gas](#)

📅 10/2014

📍 Florence, Italy

### GE EHS Orientation

[GE Oil & Gas](#)

📅 10/2013

📍 Florence, Italy

### ATEX Intermediate

[GE Oil & Gas](#)

📅 09/2014

📍 Florence, Italy

### Total Quality Lean Six Sigma Green Belt

[GE Oil & Gas](#)

📅 06/2014

📍 Florence, Italy

### PidXp and P&ID

[GE Oil & Gas](#)

📅 11/2013

📍 Florence, Italy

### Intellectual Property

[GE Oil & Gas](#)

📅 11/2013

📍 Florence, Italy

### GE Oil & Gas Product Familiarization

[GE Oil & Gas](#)

📅 07/2013

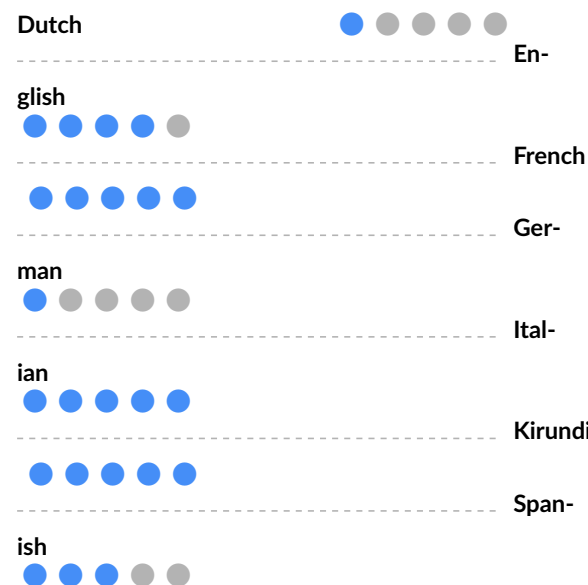
📍 Florence, Italy

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## SKILLS

Program & Project Management  
 Cross-functional Team Leadership  
 Stakeholder Engagement & Management  
 Technical Program Coordination  
 Requirements Gathering & Analysis  
 Risk Management & Mitigation  
 Budget & Resource Management  
 International Program Management  
 Technical Training & Knowledge Transfer  
 Process Optimization & Standardization  
 Vendor & Partner Management  
 Customer Relationship Management  
 Electromagnetic Simulation & Analysis  
 Finite Element Analysis  
 Testing & Validation  
 Data Acquisition Systems  
 Instrumentation Design  
 Scripting (Python, Matlab)  
 C++ & C Programming  
 Technical Documentation  
 Presentation & Facilitation

## LANGUAGES



## PUBLICATIONS

### Patents

- L. Ntibarikure, "Module with sensors arrangement," US20180180464A1, 2018. [Online]. Available: <https://patents.google.com/patent/US20180180464A1/en>

### Books

- L. Ntibarikure, *Contributions to the Art of Finite Elements in Electromagnetics*. Florence, Italy, 2014. [Online]. Available: <https://hdl.handle.net/2158/843133>

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### Journal Articles

- L. Ntibarikure, "Multiphysics simulations for 5g rfics and socs," *Bits&Chips*, 2019.
- F. Barone, A. Signorini, L. Ntibarikure, T. Fiore, F. Di Pasquale, and C. J. Oton, "Fiber-optic liquid level sensing by temperature profiling with an fbg array," *Sensors*, vol. 18, no. 8, 2018. [Online]. Available: <https://www.mdpi.com/1424-8220/18/8/2422>
- L. Ntibarikure, G. Pelosi, and S. S. and, "Harmonic balance domain decomposition finite elements for nonlinear passive microwave devices analysis," *Electromagnetics*, vol. 34, no. 3-4, pp. 239–252, 2014. DOI: 10.1080/02726343.2014.877756
- L. Ntibarikure, G. Pelosi, and S. Selleri, "Efficient harmonic balance analysis of waveguide devices with nonlinear dielectrics," *IEEE Microwave and Wireless Components Letters*, vol. 22, no. 5, pp. 221–223, 2012. DOI: 10.1109/LMWC.2012.2192420

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### Conference Proceedings

- L. Ntibarikure, G. Pelosi, and S. Selleri, "Assessment of the performances of gmres(r) using a domain decomposition approach as a preconditioner," in *Proceedings, XX Riunione Nazionale di Elettromagnetismo (XX RiNEm)*, Padua, Italy, 2014. [Online]. Available: <https://inspirehep.net/literature/1403858>
- L. Ntibarikure, "Model order reduction in finite elements analysis of phased array antennas," in *Proceedings, XIX Riunione Nazionale di Elettromagnetismo (XIX RiNEm)*, S. d. E. SiEm, Ed., Roma, Italy, 2012.
- L. Ntibarikure, G. Pelosi, and S. Selleri, "Harmonic balance finite element analysis of third order intermodulation products in ferrite devices," in *Proceedings, XIX Riunione Nazionale di Elettromagnetismo (XIX RiNEm)*, S. d. E. SiEm, Ed., Roma, Italy, 2012.